CPU Scheduling Pseudocode

FCFS

```
BEGIN
INPUT n
FOR i = 0 to n-1
    INPUT at[i]
    INPUT bt[i]
END FOR
SORT processes based on arrival time
ct[0] = at[0] + bt[0]
FOR i = 1 to n-1
    IF ct[i-1] < at[i]
        ct[i] = at[i] + bt[i]
    ELSE
        ct[i] = ct[i-1] + bt[i]
END FOR
FOR i = 0 to n-1
    tat[i] = ct[i] - at[i]
    wt[i] = tat[i] - bt[i]
END FOR
CALCULATE average TAT and WT
PRINT results
END
```

SJF

```
BEGIN
INPUT n
INPUT arrival time and burst time
WHILE all processes not completed
    SELECT process with smallest burst time
    (among arrived processes)
    IF none available
        increase time
    ELSE
        execute process completely
        calculate CT, TAT, WT
    END IF
END WHILE
CALCULATE average TAT and WT
PRINT output
END
```

Priority (Preemptive)

```
BEGIN
INPUT n
INPUT arrival time, burst time, priority
WHILE all processes not completed
    SELECT process with highest priority
    (smallest priority value among arrived)
    IF none available
```

```
        increase time
    ELSE
        execute process for 1 unit time
        reduce remaining time
        IF process finishes
            calculate CT
        END IF
END WHILE
CALCULATE TAT = CT - AT
CALCULATE WT = TAT - BT
CALCULATE average TAT and WT
PRINT output
END
```

Priority (Non-Preemptive)

```
BEGIN
INPUT n
INPUT arrival time, burst time, priority
WHILE all processes not completed
    SELECT process with highest priority
    (smallest value among arrived)
    IF none available
        increase time
    ELSE
        waiting time = current time - arrival time
        execute process completely
        turnaround time = waiting time + burst time
        mark process as completed
    END IF
END WHILE
CALCULATE average waiting time and turnaround time
PRINT output
END
```

Round Robin

```
BEGIN
INPUT n
INPUT arrival time, burst time
INPUT time quantum
SET remaining time = burst time
WHILE all processes not completed
    FOR each process
        IF process has arrived AND remaining time > 0
            IF remaining time > time quantum
                execute for time quantum
                reduce remaining time
            ELSE
                execute completely
                calculate waiting time
                mark as completed
            END IF
        END IF
    END FOR
```

```
    IF no process executed
        increase time
    END IF
END WHILE
CALCULATE turnaround time = burst + waiting
CALCULATE average waiting and turnaround time
PRINT output
END
```

SRTF

```
BEGIN
INPUT n
INPUT arrival time, burst time
SET remaining time = burst time
WHILE all processes not completed
    SELECT process with smallest remaining time
    (among arrived processes)
    IF none available
        increase time
    ELSE
        execute for 1 unit time
        reduce remaining time
        IF process finishes
            calculate completion time
        END IF
END WHILE
CALCULATE TAT = CT - AT
CALCULATE WT = TAT - BT
CALCULATE average TAT and WT
PRINT output
END
```